

User Guide

How to use the prototype: registering instruments, opening tests, recording OIML R 76 observations, closing and approving reports, and issuing certificates of conformity.

2 0 0 1 0

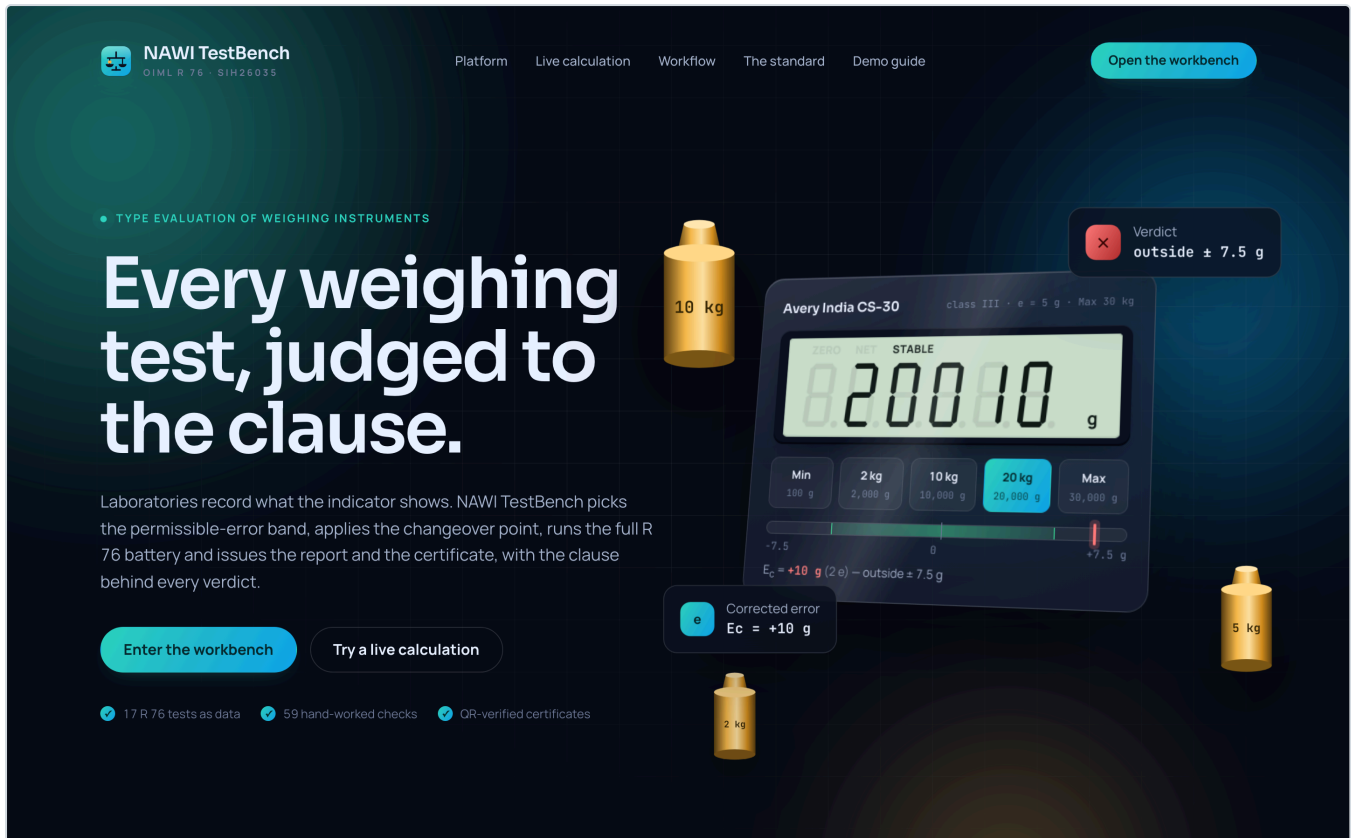
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1 What the prototype is

NAWI TestBench is a web application for laboratories that evaluate non-automatic weighing instruments (NAWIs) under OIML Recommendation R 76. The engineer records what the indicator shows; the software picks the permissible-error band for every load, applies the changeover-point correction and the zero-error correction, judges every test against its R 76 criterion, and produces the standardised report and, when everything passes, a certificate of conformity.

Everything the software knows about R 76 is stored as data in a versioned rule-set file. Every report remembers which rule set it was judged under, so a revised Recommendation becomes a new file, not new software.



The overview page. The indicator on the right is live: press a test load to see the reading, the corrected error and whether it falls inside the permissible band.

2 Starting the software

The prototype runs on any computer with Node.js 20 or later. PDF export additionally needs Google Chrome, Chromium or Microsoft Edge installed; without one, HTML and Word exports still work.

```
cd ~/dev/nawi
npm install      # first time only
npm run seed    # first time only: two worked examples and the demo accounts
npm start       # then open http://localhost:4178
```

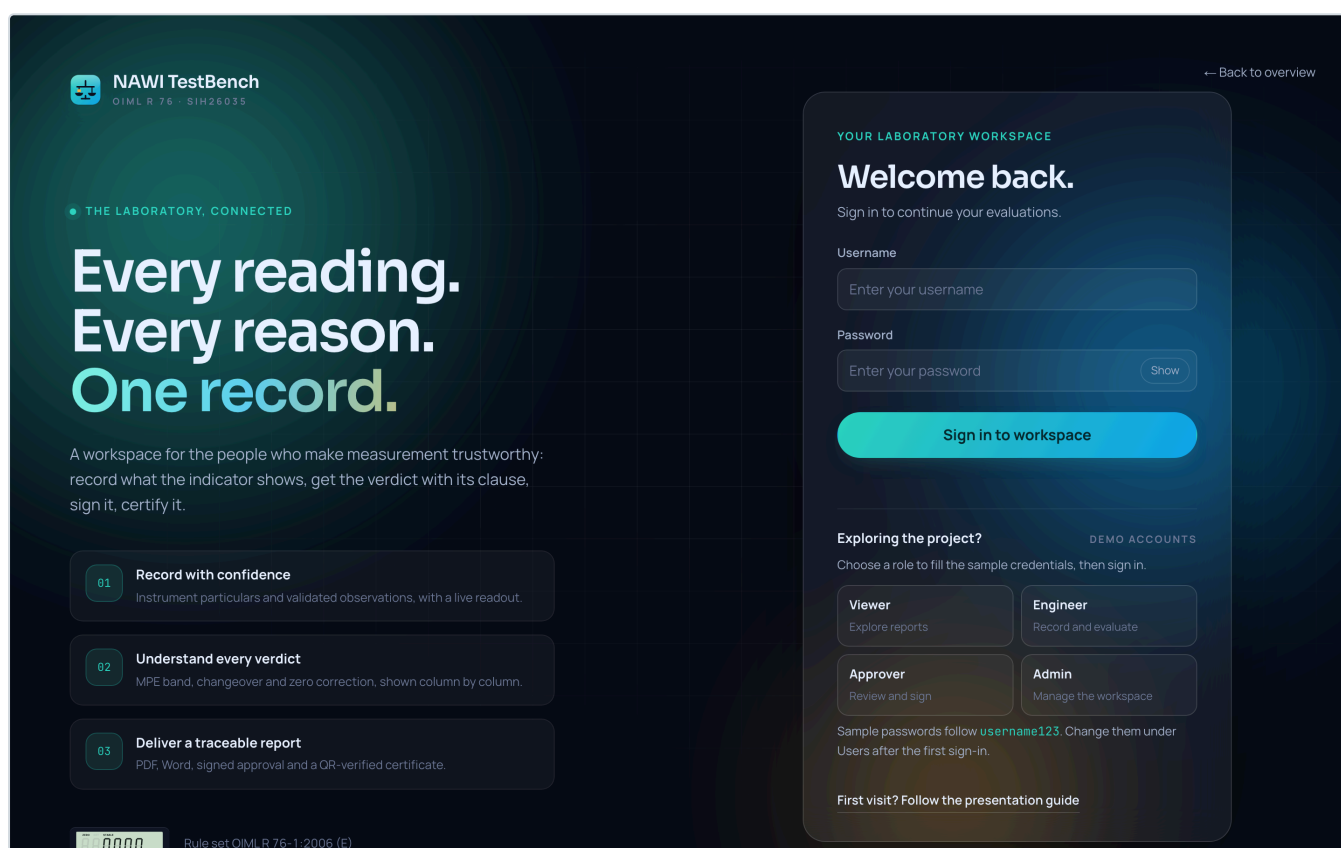
Leave the terminal window open while you use the software; closing it stops the server. To run it permanently, see `docs/deployment.md`.

If the page does not open, the server is not running. Run `npm start` again. If it reports that the port is in use, another copy is already running; stop that one first or start this one with `PORT=5000 npm start`.

3 Signing in and roles

Open `http://localhost:4178/login`. Four demonstration accounts exist after seeding. Click a username on the login page to fill the form.

Username	Password	Role	What the role can do
viewer	viewer123	Viewer	Read every test, instrument and report; download exports. Cannot change anything.
engineer	engineer123	Engineer	Everything a viewer can, plus register and edit instruments, open tests, record observations, environment readings, checklists and attachments, close and reopen reports.
approver	approver123	Approver	Everything an engineer can, plus approve and sign closed reports, return them to the engineer, withdraw approvals. Approval issues the certificate.
admin	admin123	Admin	Everything, plus the Users page: create accounts, change roles, disable accounts, reset passwords.



The sign-in page. Sessions last 12 hours; sign out from the masthead.

Change the demonstration passwords before showing the software outside the team. Sign in as admin , open Users and use “reset password” on each account.

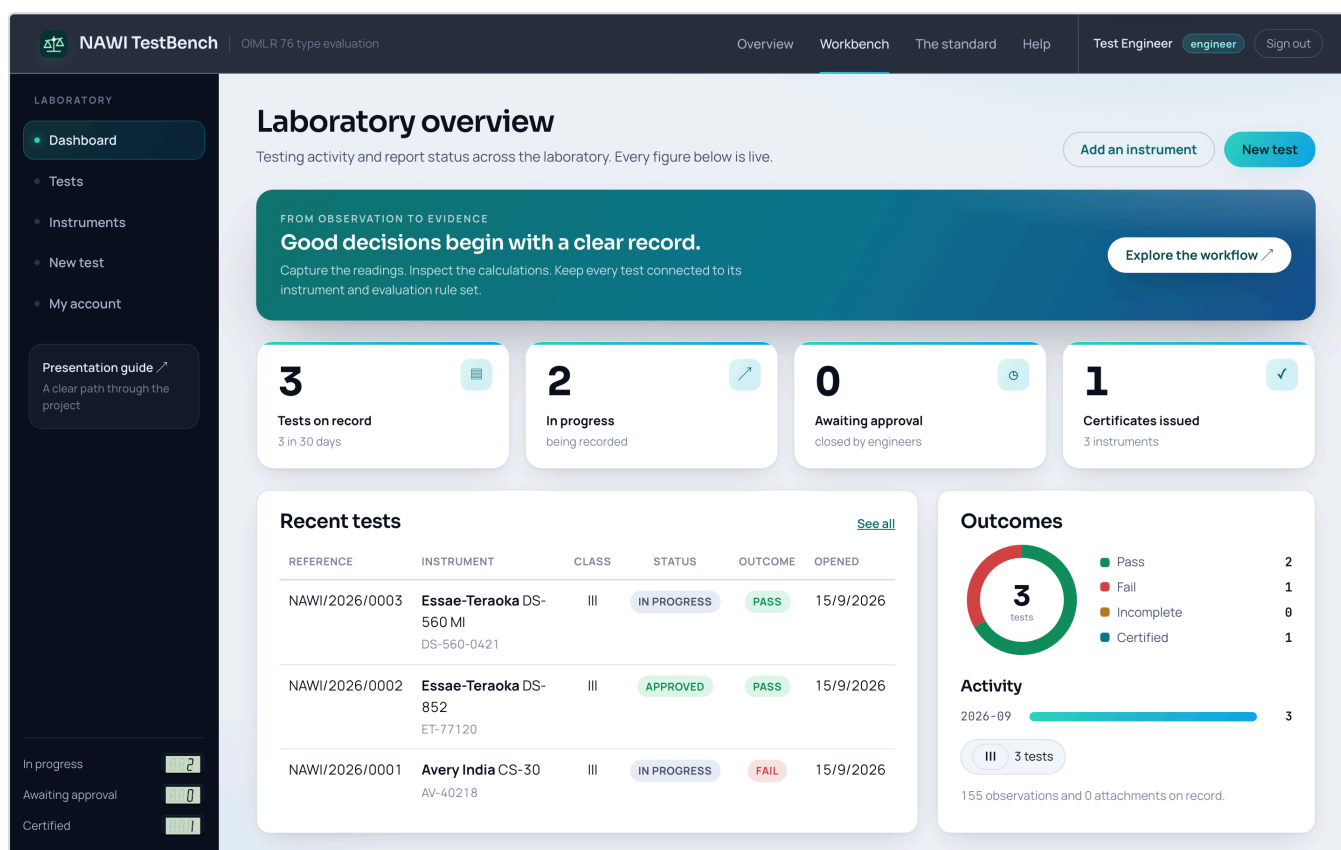
4 The four pages

Address	Page	Who	Purpose
/	Overview	Anyone	Explains the method, the battery and the workflow; interactive indicator.

Address	Page	Who	Purpose
/app	Workbench	Signed-in users	Dashboard, Tests, Instruments, New test, Users. This is where all work happens.
/standards	The standard	Anyone	Everything the engine knows about R 76, read live from the rule set, plus a permissible-error calculator.
/verify/<certificate no.>	Verify	Anyone	Confirms whether a certificate number is genuine and still valid.

Inside the workbench, the dark rail on the left switches between views. The small readouts at the bottom of the rail always show how many tests are in progress, awaiting approval and certified.

5 Dashboard



The dashboard as the engineer sees it.

- **Console readouts:** tests on record, in progress, awaiting approval, certificates issued. Click a readout to open the Tests list filtered to that status.
- **New test** and **Add an instrument** buttons (engineer and above).
- **Recent tests:** the eight most recent reports. Click a row to open it.
- **Outcomes:** how many tests currently pass, fail or are incomplete, and how many are certified. Below it, tests opened per month, per accuracy class, and totals of observations and attachments.

6 Instruments

An instrument is registered once and reused in every test made on it. Open **Instruments** in the rail and press **Add an instrument**.

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OIMLR 76 type evaluation

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engineer

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Instruments

Particulars are captured once and carried into every test the instrument appears in. Once an instrument is in a closed report its particulars are frozen.

Add an instrument

Register an instrument

Cancel

Applicant and identity

Applicant

Essae-Teraoka Pvt. Ltd.

Applicant address

Bengaluru, Karnataka

Manufacturer

Essae-Teraoka

Manufacturer address

Bengaluru, Karnataka

Model / type designation

DS-852

Serial number of EUT

ET-39014

Instrument category

Counter scale

Indication

Digital

Electronic instrument

Yes

Range type

Single range

Metrological characteristics

Accuracy class

III – Medium

Units

g

Max

15000

Min

100

Verification scale interval e

5

Actual scale interval d

5

Max additive tare T = +

7000

Max subtractive tare T = -

optional

n = 3,000 (500–10000 permitted for class III) · Min must be ≥ 100 g · changeover method required (d > 0.2 e) · MPE changes at 2,500 g, 10,000 g

Classification admissible.

Power supply and environment

Power supply

Public mains (AC)

Nominal voltage U (V)

230

Minimum operating voltage (V)

battery / DC supplies

Frequency (Hz)

50

Marked temperature limit, low (°C)

blank = -10

Marked temperature limit, high (°C)

blank = +40

Limiting value of tilting

Level indicator with ring marking

Construction

Software identification

v2.14

Load cell(s)

1 × single-point, 20 kg

Interfaces

RS-232 printer port

Notes

Save instrument

Fill a worked example

In progress2

Awaiting approval0

Certified1

The registration form after “Fill a worked example”. The grey strip under the metrological fields is the live class check.

6.1 The fields

Applicant, applicant address

The company applying for approval. Printed on the report and certificate. Leave blank if the manufacturer is the applicant.

Manufacturer, manufacturer address	Required. Printed as the manufacturer's mark.
Model / type designation, Serial number of EUT	Required. EUT is the equipment under test.
Instrument category	Counter scale, platform scale, weighbridge and so on. Descriptive only.
Indication, Electronic instrument	Digital or analogue; electronic yes or no. Warm-up, voltage and span-stability tests apply to electronic instruments only.
Range type	Single range, multi-interval or multiple range. The prototype evaluates single-range instruments fully; the others are recorded for the report.
Accuracy class	I, II, III or IIII. Decides the MPE bands and which tests apply.
Units	g or kg. Every load and reading for this instrument is entered in this unit.
Max, Min	Maximum and minimum capacity.
Verification scale interval e, Actual scale interval d	d is what the display steps in; e is the interval used for classification. Usually equal on class III instruments.
Max additive / subtractive tare	T = + and T = – from the data plate. Used to check tare entries and to suggest the eccentricity load.
Power supply, nominal voltage, minimum operating voltage, frequency	Decide the voltage limits for the voltage-variation test (3.9.3). Mains: 0.85 U to 1.10 U. External or battery supplies need the minimum operating voltage.
Marked temperature limits	Leave blank for the default –10 / +40 °C. If the instrument is marked with special limits, enter both; class III needs a span of at least 30 °C.
Limiting value of tilting	How the tilt limit is defined (level indicator, tilt sensor, or 50/1000). Printed with the tilting test.
Software identification, load cells, interfaces, notes	Printed in the general-information block of the report.

6.2 What is checked as you type

The grey strip below the metrological fields shows **n = Max / e** and whether it lies in the range Table 3 permits for the class, the lowest Min allowed, whether the changeover method is required ($d > 0.2 e$), and any findings. An instrument can be saved with findings, but findings block the certificate, so fix them if they are data-entry mistakes.

6.3 Viewing and editing

Click any row in the instrument list to open its detail panel with all particulars and its **test history**. **Edit particulars** is available until the instrument appears in a closed or approved report; after that it is frozen so old reports stay truthful. **New test on this instrument** opens the New test form with the instrument pre-selected.

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Certified1

Essae-Teraoka DS-560 MI

DS-560-0421

Edit particularsNew test on this instrument

Applicant

Essae-Teraoka Pvt. Ltd.

Bengaluru, Karnataka

Manufacturer

Essae-Teraoka

Bengaluru, Karnataka

Category

Retail scale (price computing) · digital · electronic · multi-interval

Class

III, Medium accuracy

Max / Min

15,000 g / 40 g

Partial ranges

Range 1: e = 2 g, d = 2 g, up to 6,000 g, n = 3,000

Range 2: e = 5 g, d = 5 g, up to 15,000 g, n = 3,000

Tare

T = +5,990 g / –

Temperature

–10 °C to 40 °C (default)

Supply

External supply, 12 V

test at 9.6 V and 14.4 V (3.9.3)

Software / load cell

R3.02 / 1 × single-point 20 kg

Interfaces

RS-232, USB

Test history

REFERENCE	PURPOSE	STATUS	OUTCOME	OPENED
NAWI/2026/0003	Verification	IN PROGRESS	PASS	15/9/2026

Instrument detail with derived values and test history.

7 Opening a new test

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New test

A test is opened against one instrument, one rule set and one purpose. Those never change after opening; everything else is recorded as you go.

What is being tested

Instrument

Essae-Teraoka DS-560 MI – DS-560-0421 (class III, Max 15,000 g)

Rule set

OIMLR 76-1, Edition 2006 (E)

Purpose

Type approval

MPE context

Initial verification

Zero error E₀ before tests

0

Application reference

applicant's file or letter no.

Laboratory

Laboratory

Regional Reference Standards Laboratory, ...

Test engineer

Test Engineer

Test date

15/09/2026

Environmental conditions at the start

Temperature (°C)

Relative humidity (%)

Barometric pressure (hPa)

Supply voltage (V)

Supply frequency (Hz)

Further readings can be added during the test; the report lists them all with time stamps.

Open test

The New test form.

Instrument

Pick the registered instrument.

Rule set	The edition of R 76 to judge against. Fixed once the test is opened.
Purpose	Type approval requires the full battery of 17 tests. Verification requires the six metrological tests and the markings checklist.
MPE context	Initial verification uses the Table 6 limits; In service doubles them (3.5.2).
Zero error E_0 before tests	The zero error determined at no load before testing. Every corrected error subtracts it unless a point records its own E_0 .
Application reference	The applicant's file or letter number, printed on the report.
Laboratory, Test engineer, Test date	Printed on the report and certificate. The engineer defaults to your name.
Environmental conditions at the start	Optional; saved as the first environment reading.

Press **Open test**. A reference like `NAWI/2026/0003` is assigned and the report workspace opens.

8 The report workspace

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- Instruments
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NAWI/2026/0001, opened 15/9/2026

Avery India CS-30 AV-40218

IN PROGRESS **FAIL** Type approval against OIML R 76-1, Edition 2006 (E), initial-verification limits

All tests **Report** Word CSV Import CSV PDF

Error envelope, class III, initial verification – every load-based point

±0.5e ±1e ±1.5e Weighing performance: 20 kg

load, in verification scale intervals

Does not conform

Weighing performance outside the permissible limits

6 of 17 tests required for type approval decided. Outstanding:

- Tilting
- Warm-up time
- Static temperatures
- Temperature effect on no-load indication
- Power supply voltage variation
- Damp heat, steady state
- Creep
- Zero return
- Span stability
- Functional and construction checks
- Repeatability (incomplete)

In progress 2

Awaiting approval 0

Certified 1

Evaluation fails. Closing freezes the observations and sends the report for approval. **Close report**

6 of 17 required tests decided

Top of a report in progress. The 20 kg point of this worked example is outside the permissible error, so the outcome reads FAIL.

From top to bottom:

1. **Header:** reference, dates, instrument, status badges, and the buttons **All tests**, **Report** (HTML in a new tab), **Word**, **PDF** and, once issued, **Certificate**.

2. **Error envelope:** every load-based point plotted against the MPE staircase for the class. Black dots are within the limit, red dots outside; hover a dot for its values.
3. **Outcome panel:** the LCD reads PASS, FAIL or ---- (incomplete), with DRAFT, CLOSED, APPROVED and CERT annunciators. Below it, what the purpose still requires and whether the report is eligible for a certificate.
4. **Workflow box:** the action available at this stage (close, approve, return, reopen). See section 14.
5. **Instrument and Evaluation particulars.**
6. **Laboratory environment** readings with an add form (section 12).
7. **Test groups:** Metrological performance, Influence factors, Time and stability, Construction and markings. Each test is a collapsible block with its clause, its verdict badge and its observation count. A coloured left edge repeats the verdict.
8. **Photographs and documents, Remarks, Audit trail.**
9. **Navigator** (right column, stays in view while scrolling): progress bar and a link to every test with a verdict dot. Click a test to jump to it and open it.

Everything is recalculated every time an observation is saved or removed. There is no “calculate” button.

9 Recording observations and the live readout

Open a test block and use the **Record an observation** form at its bottom. Fields common to the load-based tests:

Point / Position / Run	A label for the report: “10 kg”, “Segment 2”, “Run 3”. Eccentricity offers the four segments; repeatability suggests the next run number.
Direction	Increasing or decreasing load (A.4.4.1 asks for both).
Load	The test load applied, in the instrument's unit.
Tare	Tare test only: the tare value in use. The net load is judged.
Indication	What the display shows.
Added until display stepped, ΔL	Changeover-point method: the small weights added until the display moved up one interval. Leave blank to use the raw reading. Required in spirit when $d > 0.2 e$.
Zero error E_0 at this point	Overrides the session E_0 for this point.
Remark	Free text printed under the row in the report.

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In progress

2

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0

Certified

1

25 kg

↓

25,000 g

25,005 g

2.5 g

25,005 g

+5

0

+5

± 7.5 g

2000-10000e

PA

30 kg (Max)

↓

30,000 g

30,000 g

2.5 g

30,000 g

0

0

0

± 7.5 g

2000-10000e

PA

15 kg

↑

15,000 g

15,005 g

3.5 g

15,004 g

+4

0

+4

± 7.5 g

2000-10000e

PA

Load plan (A.4.4.1) satisfied: 10 loads including Min, Max and the band changes at 2,500, 10,000.

RECORD AN OBSERVATION

Point

Direction

Load (g)

20 kg

Increasing

20000

Indication (g)

Added until display stepped, ΔL (g)

Zero error E_0 at this point (g)

20010

2.5

blank = sessio

Remark

optional

Include Min (100 g), Max (30,000 g) and the band changes at 2,500 g, 10,000 g; at least 10 loads, increasing then decreasing.

Record

> Eccentricity

PASS

R 76-1 3.6.2 / A.4.7

4 observations

> Repeatability

INCOMPLETE

R 76-1 3.6.1 / A.4.10

3 observations

> Discrimination

PASS

R 76-1 3.8.2.2 / A.4.8.2

1 observation

6 of 17 required tests decided

METROLOGICAL PERFORMANCE

Weighing performance

Eccentricity

Repeatability

Discrimination

Tare device

Zero-setting accuracy

INFLUENCE FACTORS

Tilting

Warm-up time

Static temperatures

Temperature effect on no-load indication

Power supply voltage variation

Damp heat, steady state

TIME AND STABILITY

Creep

Zero return

Span stability

CONSTRUCTION AND MARKINGS

Descriptive markings

Functional and construction

As you type, the readout on the right shows the indication, the corrected error on a gauge against $\pm$ MPE, and the arithmetic. Here 20 010 g at a 20 kg load gives +10 g against $\pm$ 7.5 g: outside.

9.1 What happens when you press Record

- Errors** stop the save and are shown under the form: load above Max (+ T), negative load, tare above T, missing second indication, missing time, a creep reading at a different load, a wrong tilt position.
- Warnings** let the save go through but are stored with the observation, shown as $\triangle$ next to the point, listed under the table and printed in the report: indication not a multiple of d, ΔL larger than one e, E_0 larger than one e, load below Min, a temperature or voltage outside the instrument's limits, a duplicate label.
- The table, the envelope, the outcome panel and the navigator update at once.

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Test loads applied from zero up to Max and removed back to zero. At least 10 loads for the initial intrinsic error, 5 otherwise, including Min, Max and the loads at which the mpe changes. · |error| ≤ MPE for the applied load

POINT	↓ ↑	LOAD	INDICATION	ADDED ΔL	CORRECTED	E	E ₀	EC	MPE	BAND	RES
Min	↓	100 g	100 g	1.5 g	101 g	+1	0	+1	± 2.5 g	0-500e	PA
2 kg	↓	2,000 g	2,000 g	1.5 g	2,001 g	+1	0	+1	± 2.5 g	0-500e	PA
2.5 kg (band change)	↓	2,500 g	2,500 g	2 g	2,500.5 g	+0.5	0	+0.5	± 2.5 g	0-500e	PA
5 kg	↓	5,000 g	5,000 g	1 g	5,001.5 g	+1.5	0	+1.5	± 5 g	500-2000e	PA
10 kg (band change)	↓	10,000 g	10,000 g	2 g	10,000.5 g	+0.5	0	+0.5	± 5 g	500-2000e	PA
15 kg	↓	15,000 g	15,005 g	3 g	15,004.5 g	+4.5	0	+4.5	± 7.5 g	2000-10000e	PA
20 kg	↓	20,000 g	20,010 g	—	20,010 g	+10	0	+10	± 7.5 g	2000-10000e	Ff
25 kg	↓	25,000 g	25,005 g	2.5 g	25,005 g	+5	0	+5	± 7.5 g	2000-10000e	PA
30 kg (Max)	↓	30,000 g	30,000 g	2.5 g	30,000 g	0	0	0	± 7.5 g	2000-10000e	PA
15 kg	↑	15,000 g	15,005 g	3.5 g	15,004 g	+4	0	+4	± 7.5 g	2000-10000e	PA

Load plan (A.4.4.1) satisfied: 10 loads including Min, Max and the band changes at 2,500, 10,000.

RECORD AN OBSERVATION

Point

Direction

Load (g)

Indication (g)

Added until display stepped. ΔL (g)

Zero error E₀ at this point (g)

Type a load and an indication to see the corrected error against the permissible limit before you record it.

6 of 17 required tests decided

METROLOGICAL PERFORMANCE

- Weighing performance
- Eccentricity
- Repeatability
- Discrimination
- Tare device
- Zero-setting accuracy

INFLUENCE FACTORS

- Tilting
- Warm-up time
- Static temperatures
- Temperature effect on no-load indication
- Power supply voltage variation
- Damp heat, steady state

TIME AND STABILITY

- Creep
- Zero return
- Span stability

CONSTRUCTION AND MARKINGS

- Descriptive markings
- Functional and construction

A weighing table: I, ΔL, corrected indication, E, E₀, Ec, MPE, band and verdict for every point. “remove” deletes a point while the report is open.

9.2 The arithmetic behind each row

$P = I + \frac{1}{2}e - \Delta L$ indication before rounding (blank ΔL → P = I)
 $E = P - L$ error of indication
 $E_c = E - E_0$ corrected error, compared with the MPE for L
 MPE from Table 6: ±0.5 e, ±1 e or ±1.5 e depending on L / e and the class; × 2 in service

10 Test-by-test instructions

Each test block explains its criterion and, where the standard prescribes them, the loads or sequence to use. This table summarises what to enter.

Test (clause)	Applies to	What to record	How it is judged
Weighing performance (3.5.1, A.4.4.1)	All	At least 10 loads (type approval) or 5 (verification) from zero to Max and back, including Min, Max and the loads where the MPE changes. The form lists those loads for this instrument.	Ec ≤ MPE for every point. The load plan under the table says what is still missing.

Test (clause)	Applies to	What to record	How it is judged
Eccentricity (3.6.2, A.4.7)	All	One third of (Max + T) on each of the four segments. Choose the segment from the list.	$ Ec \leq MPE$ at each segment.
Repeatability (3.6.1, A.4.10)	All	Type approval: two series, about 50 % of Max and near Max, 10 weighings each when Max < 1000 kg. Verification: one series near 0.8 Max, 3 runs (III, IIII) or 6 (I, II).	Spread of the series $\leq MPE$ at that load; incomplete until the run count is met.
Discrimination (3.8.2.2)	$d \geq 5 \text{ mg}$	Load, indication before, indication after gently adding 1.4 d.	Indication must change by at least 1 d.
Tare device (3.5.3.3, 4.6.3)	All	Gross load, tare value, indication.	Net error $\leq MPE$ for the net load.
Zero-setting accuracy (4.5.2)	All	Indication after zero setting at nominal 0.	$ \text{error} \leq 0.25 \text{ e}$.
Tilting (3.9.1.1, A.5.1)	II, III, IIII	Record the no-load indication in the reference (level) position first. Then in each tilted position: no load, self-indication capacity and Max.	No-load shift $\leq 2 \text{ e}$ (class II exempt); loaded $ Ec \leq MPE$.
Warm-up time (5.3.5, A.5.2)	Electronic	Load near Max at 0, 5, 15 and 30 min after switch-on, each with its own E_0 .	$ Ec \leq MPE$ at every time; incomplete until all four are recorded.
Static temperatures (3.9.2, A.5.3.1)	All	Weighing points with the chamber temperature as the condition: reference, high, low, 5 °C if low ≤ 0 °C, reference. The form suggests the temperatures for this instrument.	$ Ec \leq MPE$ under every condition; at least four conditions.
Temperature effect on no-load (3.9.2.3)	All	Zero indication at each temperature, in the order tested.	Change $\leq 1 \text{ e}$ per 5 °C (per 1 °C for class I) between consecutive temperatures.
Voltage variation (3.9.3, A.5.4)	Electronic	Points at 10 e and between $\frac{1}{2}$ Max and Max, with the supply voltage as the condition: lower limit, nominal, upper limit. The limits are computed from the instrument's supply type.	$ Ec \leq MPE$ at all three conditions.

Test (clause)	Applies to	What to record	How it is judged
Damp heat (B.2)	Not class I; not class II with $e < 1$ g	At least five loads under each of: reference 50 % RH, high temperature 85 % RH, reference again.	$ E_c \leq \text{MPE}$ everywhere; three conditions.
Creep (3.9.4.1, A.4.11.1)	II, III, IIII	Load near Max; indication on loading, at 15 and 30 min. Continue to 4 h only if the 30-minute limits are missed.	Drift $\leq 0.5 e$ in 30 min and $\leq 0.2 e$ between minute 15 and 30; otherwise drift over 4 h $\leq \text{MPE}$.
Zero return (3.9.4.2)	II, III, IIII	Zero before and after 30 min under a load near Max.	Deviation $\leq 0.5 e$.
Span stability (5.3.3, B.4)	Electronic, not class I	At least eight measurements near Max spread over up to 28 days, same weights. Use the date as the label.	Each $ E_c \leq \text{MPE}$; max – min of the errors $\leq \max(0.5 e, 0.5 \text{ MPE})$.
Descriptive markings (7.1)	All	Checklist, see section 11.	Every compulsory item present.
Functional checks (4, 5.3)	Type approval	Checklist, see section 11.	Every applicable item satisfied.

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Creep

A load close to Max stays on the instrument. The indication may drift by at most 0.5 e during the first 30 minutes and by at most 0.2 e between minute 15 and minute 30; failing that, the drift over four hours must stay within the mpe.

No observations recorded for this test.

RECORD A CREEP READING

Reading: On loading | Minutes after loading: 0 | Load (g): 30000

Indication (g): | Readings at 0, 15 and 30 min decide the test; continue to 4 h only if the 30-minute limits are missed.

Remark: optional

Record

6 of 17 required tests decided

METROLOGICAL PERFORMANCE

- Weighing performance
- Eccentricity
- Repeatability
- Discrimination
- Tare device
- Zero-setting accuracy

INFLUENCE FACTORS

- Tilting
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TIME AND STABILITY

> Zero return NOT PERFORMED | R 76-1 3.9.4.2 / A.4.11.2 | 0 observations

The creep form pre-fills the next time step and keeps the load of the series.

Tests that do not apply to the instrument (for example creep on a class I balance) show **N/A** with the reason and need no entry.

11 Checklists

NAWI TestBench

OIMLR 76 type evaluation

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Awaiting approval

0

Certified

1

Presence and form of the descriptive markings on the instrument or its data plate. · every compulsory item present; every applicable item present

#	REQUIREMENT	CLAUSE	LEVEL	FINDING	REMARK
1	Manufacturer's mark or full name	7.1.1	Compulsory	Yes No N/A	optional
2	Accuracy class as a Roman numeral in an oval	7.1.1	Compulsory	Yes No N/A	optional
3	Maximum capacity in the form Max ...	7.1.1	Compulsory	Yes No N/A	optional
4	Minimum capacity in the form Min ...	7.1.1	Compulsory	Yes No N/A	optional
5	Verification scale interval in the form e = ...	7.1.1	Compulsory	Yes No N/A	optional
6	Serial number	7.1.2	If applicable	Yes No N/A	optional
7	Name or mark of the manufacturer's agent (imported)	7.1.2	If applicable	Yes No N/A	optional
8	Software identification (software-controlled instruments)	7.1.2	If applicable	Yes No N/A	optional
9	Actual scale interval in the form d = ... (when d < e)	7.1.2	If applicable	Yes No N/A	optional
10	Maximum additive tare effect in the form T = + ...	7.1.2	If applicable	Yes No N/A	optional
11	Maximum subtractive tare effect in the form T = - ...	7.1.2	If applicable	Yes No N/A	optional

6 of 17 required tests decided

METROLOGICAL PERFORMANCE

Weighing performance

Eccentricity

Repeatability

Discrimination

Tare device

Zero-setting accuracy

INFLUENCE FACTORS

Tilting

Warm-up time

Static temperatures

Temperature effect on no-load indication

Power supply voltage variation

Damp heat, steady state

TIME AND STABILITY

Creep

Zero return

Span stability

CONSTRUCTION AND MARKINGS

Descriptive markings

Functional and construction

The descriptive-markings checklist. Choices save immediately.

The two construction tests are inspections rather than measurements. For every item choose **Yes**, **No** or **N/A** and add a remark if useful. Rules:

- A **compulsory** item (manufacturer's mark, class oval, Max, Min, e, legibility) must be Yes. No or N/A fails the test.
- An **if applicable** item (serial number, software identification, T = +, special temperature limits and so on) may be N/A when the feature is absent. No fails the test.
- The test stays **INCOMPLETE** until every compulsory item is answered.

12 Laboratory environment

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STAGE	TEMPERATURE (°C)	RH (%)	PRESSURE (hPa)	VOLTAGE (V)	FREQUENCY (Hz)	NOTE	RECORDED
Start of test	24.5	48	1,008	231	50		15 Sept 2026, 2:05 pm
Stage	T (°C)	RH (%)	p (hPa)	U (V)	f (Hz)	Note	
During test							

Metrological performance

5 of 6 decided

Weighing performance

FAIL

R 76-1 3.5.1 / A.4.4.1

10 observations

Test loads applied from zero up to Max and removed back to zero. At least 10 loads for the initial intrinsic error, 5 otherwise, including Min, Max and the loads at which the mpe changes. · |error| ≤ MPE for the applied load

POINT	↓↑	LOAD	INDICATION	ADDED ΔL	CORRECTED	E	E ₀	EC	MPE	BAND	RES
-------	----	------	------------	----------	-----------	---	----------------	----	-----	------	-----

6 of 17 required tests decided

METROLOGICAL PERFORMANCE

Weighing performance

Eccentricity

Repeatability

Discrimination

Tare device

Zero-setting accuracy

INFLUENCE FACTORS

Tilting

Warm-up time

Static temperatures

Environment readings with the inline add form.

Record temperature (°C), relative humidity (%), barometric pressure (hPa), supply voltage (V) and frequency (Hz) with a stage label such as “Start of test”, “During test”, “End of test” or “Temperature chamber”. Each reading is time-stamped. A reading outside the instrument's temperature limits or supply limits is saved with a warning. All readings are printed in section 2 of the report.

13 Photographs, documents and remarks

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0

Certified

1

> Descriptive markings

PASS

R 76-1 7.1

14 of 14 items

> Functional and construction checks

NOT PERFORMED

R 76-1 4 / 5.3

0 of 0 items

Photographs and supporting documents

Data plate, markings, set-up, manufacturer's documents

Nothing attached yet.

Drop images or PDFs here, or **choose files**. Up to 8 MB each.

Remarks

Saved when you leave the field

Observations about the instrument, deviations from the procedure, conditions worth recording...

Audit trail

15 Sept 2026, 2:05 pm F. Shahid report.open: NAWI/2026/0001 for Avery India CS-30

6 of 17 required tests decided

METROLOGICAL PERFORMANCE

Weighing performance

Eccentricity

Repeatability

Discrimination

Tare device

Zero-setting accuracy

INFLUENCE FACTORS

Tilting

Warm-up time

Static temperatures

Temperature effect on no-load indication

Power supply voltage variation

Damp heat, steady state

TIME AND STABILITY

Creep

Zero return

Drop zone for photographs and documents; remarks save when you leave the field.

- Drop images (JPEG, PNG, WebP), PDFs, text, CSV, Word or Excel files onto the drop zone, or click **choose files**. Up to 8 MB each. You are asked for a caption. Photographs appear in the PDF and HTML report; other documents are listed with type and size.

- **Remarks** are free text for the report: instrument condition on arrival, deviations from procedure, test weights used.
- The **audit trail** lists every action on the report with who did it and when.

14 Closing, approval and certificates

```
draft --close--> closed (awaiting approval) --approve--> approved (+ certificate if eligible)
^          |                                     |
+----- reopen -----+----- return to engineer -----+----- withdraw and reopen (approver)
```

1. **Engineer closes.** Press **Close report** in the workflow box. Observations are frozen. If the battery is incomplete you are asked to confirm; the report is then filed as incomplete and cannot be certified.
2. **Approver reviews.** Sign in as an approver and open the report. The workflow box offers an optional note, **Return to engineer** (reopens it with the note) and **Approve and sign**.
3. **Approval signs the report.** A SHA-256 signature is computed over the instrument, every observation and check, every verdict and the approval record, and printed on the report. If the evaluation passed, every test required for the purpose is decided and the instrument has no classification findings, a **certificate number** like `NAWI-C/2026/0001` is issued and the **Certificate** button appears.
4. **Verification.** Anyone can open `/verify/<certificate number>`, or scan the QR code printed on the certificate and the report, to confirm that the certificate exists and is current. Withdrawing the approval marks it as withdrawn.

NAWI TestBench | OIMLR 76 type evaluation | Overview | Workbench | The standard | Help | Approving Officer **approver** | Sign out

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Awaiting approval 1

Certified 1

NAWI/2026/0001, opened 15/9/2026, closed 15/9/2026

Avery India CS-30 AV-40218

AWAITING APPROVAL **FAIL** Type approval against OIMLR 76-1, Edition 2006 (E), initial-verification limits

All tests **Report** Word CSV PDF

Error envelope, class III, initial verification – every load-based point

Graph showing error envelope (load in verification scale intervals) with a red dot indicating 'Weighing performance: 20 kg'.

FAIL

Does not conform

Weighing performance outside the permissible limits

6 of 17 tests required for type approval decided. Outstanding:

- Tilting
- Warm-up time
- Static temperatures
- Temperature effect on no-load indication
- Power supply voltage variation
- Damp heat, steady state
- Creep
- Zero return
- Span stability
- Functional and construction checks
- Repeatability (incomplete)

Closed by the test engineer with outcome fail. Approving will sign the report; no certificate is issued for this outcome.

Reopen Approval note (optional) Return to engineer **Approve and sign**

Instrument Evaluation

6 of 17 required tests decided

A closed report as the approver sees it: note field, Return to engineer, Approve and sign.

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Awaiting approval

0

Certified

1

NAWI/2026/0002, opened 15/9/2026, closed 15/9/2026, approved 15/9/2026 by Approving Officer

Essae-Teraoka DS-852

ET-77120

APPROVED PASS CERTIFICATE NAWI-C/2026/0001

Type approval against OIMLR 76-1, Edition 2006 (E), initial-verification limits

All tests

Report

Word

CSV

PDF

Certificate

Error envelope, class III, initial verification – every load-based point

APPROVED

PASS

Conforms to every requirement evaluated

Every recorded point within the permissible error; battery complete

All 17 tests required for type approval are complete.

Approved and signed by Approving Officer on 15/9/2026. Certificate NAWI-C/2026/0001 issued.

Withdraw and reopen

Instrument

Applicant

Essae-Teraoka Pvt. Ltd.

Category

Platform scale · digital, electronic

Evaluation

Laboratory

Regional Reference Standards Laboratory, Delhi

Test engineer

F. Shahid

17 of 17 required tests decided

METROLOGICAL PERFORMANCE

Weighing performance

Eccentricity

An approved, certified report. The Certificate button downloads the PDF; “Withdraw and reopen” is limited to approvers.

NAWI TestBench

certificate verification

Certificate is valid

Certificate

NAWI-C/2026/0001

Report

NAWI/2026/0002

Instrument

Essae-Teraoka DS-852, serial ET-77120

Class

III · Max 15000 g · e = 5 g

Outcome

pass

Approved

Approving Officer · 15 September 2026

Signature

0f466bfa28b8d0a81ded3ddf3d2aafadd17e933b760caad769d130b13b20d651

The public verification page.

15 Reports and exports

Button / address	Format	Use
Report	HTML in a new tab	Quick review or printing from the browser.
CSV / Import CSV	.csv	Export every observation, or import a spreadsheet in the template layout through the same validation as the forms.

Button / address	Format	Use
PDF	PDF, A4	The standardised test report for the file and for the applicant. Needs Chrome/Chromium on the server; otherwise the HTML opens instead.
Word	.docx	Editable copy of the same report for laboratories that add their own cover pages.
Certificate	PDF, one page	Certificate of conformity, only after approval of a passing report.
/api/sessions/<id>/report.json	JSON	Raw observations, checks, environment and the full evaluation, for integration.

The report follows the R 76-2 structure: general information concerning the type, evaluation particulars and laboratory conditions, summary of results, test results by group with every column of the arithmetic, remarks, photographs, signatures, digital signature and audit trail.


Essae-Teraoka DS-852
NAWI/2026/0002 - Certificate NAWI-C/2026/0001

PASS

Conforms to the requirements evaluated

TYPE EVALUATION TEST REPORT - NON-AUTOMATIC WEIGHING INSTRUMENT - OIML R 76
Regional Reference Standards Laboratory, Delhi
Report generated 15 Sept 2026, 2:43 pm
Rule set OIML R 76-1, Edition 2006 (E)
Approved 15 September 2026 by Approving Officer

1 General information concerning the type

APPLICANT AND MANUFACTURER		METROLOGICAL CHARACTERISTICS	
Applicant	Essae-Teraoka Pvt. Ltd.	Accuracy class	III – Medium accuracy
Address	Bengaluru, Karnataka	Max / Min	15,000 g / 100 g
Manufacturer	Essae-Teraoka	e / d	5 g / 5 g
Address	Bengaluru, Karnataka	n = Max / e	3,000
Model / type designation	DS-852	Min lower limit	100 g
Serial number of EUT	ET-77128	Rounding elimination	required (d > 0.2 e)
Application reference	LM/TA/2026/104	Tare, additive / subtractive	T = +7,000 g / –
Instrument category	Platform scale	Temperature limits	-10 °C / 40 °C (default, 3.9.2.1)
Indication	digital, electronic	Power supply	Public mains (AC), 230 V 50 Hz
Range type	Single range	Voltage test limits (3.9.3)	195.5 V – 253 V
Software identification	v2.14	Limiting value of tilting	Level indicator with ring marking
Load cell(s)	1 × single-point 20 kg		
Interfaces	RS-232 printer port		

2 Evaluation particulars and laboratory conditions

EVALUATION		ENVIRONMENTAL CONDITIONS	
Laboratory	Regional Reference Standards Laboratory, Delhi	Stage	T (°C) RH (%) p (hPa) U (m) f (Hz) Time
Test engineer	F. Shahid	Start of test	22.8 51 1,006 229 50 15 Sept 2026, 2:05 pm
Purpose	Type approval	During test	23.1 50 1,006 230 50 15 Sept 2026, 2:05 pm
MPE context	Initial verification	End of test	23.4 49 1,005 231 50 15 Sept 2026, 2:05 pm
Zero error E ₀ (session)	0 g		
Test date	18 August 2026		
Report opened / closed	15 September 2026 / 15 September 2026		
Rule set applied	OIML R 76-1, Edition 2006 (E)		

3 Summary of results

#	Test	Clause	Observations	Result
METROLOGICAL PERFORMANCE				
1	Weighing performance	3.5.1 / A.4.4.1	14	PASS
2	Eccentricity	3.6.2 / A.4.7	4	PASS
3	Repeatability	3.6.1 / A.4.18	20	PASS
4	Discrimination	3.8.2.2 / A.4.8.2	2	PASS
5	Tare device	3.5.3.3 / 4.6.3 / A.4.6	2	PASS
6	Zero-setting accuracy	4.5.2 / A.4.2.3	1	PASS
INFLUENCE FACTORS				
1	Tilting	3.9.1.1 / A.5.1	7	PASS

First page of the generated report.


Certificate of Conformity
Regional Reference Standards Laboratory, Delhi
Type evaluation of a non-automatic weighing instrument - OIML R 76

Certificate number
NAWI-C/2026/0001
Report NAWI/2026/0002
Issued 15 September 2026

This is to certify that the instrument described below was evaluated in accordance with **OIML R 76-1, Edition 2006 (E)** and was found to conform to the metrological and technical requirements applicable to accuracy class III (Medium accuracy) for the purpose of type evaluation.

INSTRUMENT

Applicant	Essae-Teraoka Pvt. Ltd., Bengaluru, Karnataka	Manufacturer	Essae-Teraoka, Bengaluru, Karnataka
Model / type	DS-852	Serial number of EUT	ET-77128
Category	Platform scale, digital indication	Accuracy class	III – Medium accuracy
Max / Min	15,000 g / 100 g	e / d / n	5 g / 5 g / 3,000
Temperature limits	-10 °C / 40 °C	Power supply	Public mains (AC), 230 V
Software identification	v2.14		

REQUIREMENTS EVALUATED

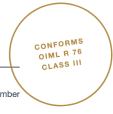
Weighing performance	3.5.1 / A.4.4.1	PASS	Eccentricity	3.6.2 / A.4.7	PASS
Repeatability	3.6.1 / A.4.18	PASS	Discrimination	3.8.2.2 / A.4.8.2	PASS
Tare device	3.5.3.3 / 4.6.3 / A.4.6	PASS	Zero-setting accuracy	4.5.2 / A.4.2.3	PASS
Tilting	3.9.1.1 / A.5.1	PASS	Warm-up time	5.3.5 / A.5.2	PASS
Static temperatures	3.9.2 / A.5.3.1	PASS	Temperature effect on no-load indication	3.9.2.3 / A.5.3.2	PASS
Power supply voltage variation	3.9.3 / A.5.4	PASS	Damp heat, steady state	3.9.5 / B.2	PASS
Creep	3.9.4.1 / A.4.11.1	PASS	Zero return	3.9.4.2 / A.4.11.2	PASS
Span stability	5.3.3 / B.4	PASS	Descriptive markings	7.1	PASS
Functional and construction checks	4 / 5.3	PASS			

The results recorded in test report NAWI/2026/0002, closed on 15 September 2026 and approved on 15 September 2026, show every error of indication within the maximum permissible error for its load and every applicable construction requirement satisfied.
Reviewed against R 76-1:2006; no deviations.



F. Shahid
Test engineer

Approving Officer
Approving officer - 15 September 2026



Scan to verify
This certificate is valid only together with test report NAWI/2026/0002 and refers solely to the instrument identified above. Digital signature (SPA-256): 8f46617a22b0a0a5d5e046f502a0fa0517e0133b708c0a070560386330c980531
Verify authenticity at <http://localhost:4178/verify/NAWI-C/2026/0001>

The certificate of conformity.

16 Finding tests again

NAWI TestBench | OIMLR 76 type evaluation

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Awaiting approval: 0
Certified: 1

Tests

Every test report, evaluated live against the rule set it was opened under. Scroll the list or narrow it down.

New test

Search reference, certificate, applicant Any sta Any out Any cla Any pur dd/mm/yyyy dd/mm/yyyy **Apply** Clear

Export filtered register - CSV

3 tests on record.

REFERENCE	INSTRUMENT	CLASS	PURPOSE	STATUS	OUTCOME	CERTIFICATE	OPENED
NAWI/2026/0003 4 of 7 required tests	Essae-Teraoka DS-560 MI DS-560-0421, Essae-Teraoka Pvt. Ltd.	III	Verification initial	IN PROGRESS	PASS	—	15/9/2026
NAWI/2026/0002 17 of 17 required tests	Essae-Teraoka DS-852 ET-77120, Essae-Teraoka Pvt. Ltd.	III	Type approval initial	APPROVED	PASS	NAWI-C/2026/0001	15/9/2026
NAWI/2026/0001 6 of 17 required tests	Avery India CS-30 AV-40218, Avery India Ltd.	III	Type approval initial	IN PROGRESS	FAIL	—	15/9/2026

The Tests view. Every report is listed; scroll, or narrow the list.

- **Search** matches the reference, certificate number, applicant, manufacturer, model, serial number and application reference.
- **Filters:** status (in progress, awaiting approval, approved), outcome (pass, fail, incomplete), accuracy class, purpose, and opened-from and opened-to dates. Selects apply immediately; press **Apply** after typing a search, **Clear** to reset.
- Each row shows a progress bar of required tests decided, status, outcome, certificate number and opening date. Click a row to open the report.
- Per-instrument history is on the instrument's detail panel.

17 Users and passwords

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Users

Roles are cumulative: a viewer reads, an engineer records and closes, an approver signs and issues certificates, an admin manages users.

Add a user

Username

Full name

RoleEngineer

Password (8+ characters)

Create user

USERNAME	NAME	ROLE	ACTIVE	SINCE	
admin	Administrator	admin	ACTIVE	15/9/2026	disable · reset_password
engineer	Test Engineer	engineer	ACTIVE	15/9/2026	disable · reset_password
approver	Approving Officer	approver	ACTIVE	15/9/2026	disable · reset_password
viewer	Read-only Viewer	viewer	ACTIVE	15/9/2026	disable · reset_password

The Users view (admin only).

Admins create accounts with a username, full name, role and a password of at least eight characters, change a role from the drop-down, disable or enable an account, and reset passwords. Passwords are stored as script hashes; sessions are HttpOnly cookies that expire after 12 hours.

18 The standard page

NAWI TestBench | OIML R 76 type evaluation

OverviewWorkbenchThe standard

OIML R 76–1, Edition 2006 (E)

Everything the evaluation engine knows about OIML R 76 is on this page, read live from the rule-set file. Work out the permissible error for any instrument and load, then see where it sits on the envelope.

Permissible error for a load

Accuracy class

III – Medium accuracy

Max (g) e (g)

30000 5

Load applied (g)

20000

Context

Initial verification

± 7.5 g 1.5 e

Load in intervals	4,000 e
Band	> 2,000 to 10,000 e
n = Max / e	6,000 (500 – 10,000 permitted)
Multiplier	× 1, initial verification

Envelope for class III, Max 30,000 g, e = 5 g

The standard page reads the rule set live.

Use the calculator at the top to find the permissible error for any class, Max, e, load and context, and see where the load sits on the envelope. Below it are the accuracy-class table, the MPE bands, and every test with its rule, clause and whether type approval or verification requires it. This is the quickest way to answer “why did that point fail?”.

19 Worked examples

NAWI/2026/0001, Avery India CS-30 (fails)

Class III counter scale, Max 30 kg, e = d = 5 g, initial verification. At 20 kg (4000 e, band above 2000 e) the MPE is 1.5 e = 7.5 g. The display reads 20 010 g with no changeover recorded, so $E_c = +10$ g and the point fails; the whole evaluation therefore fails while the other nine weighing points, eccentricity, discrimination, tare, zero and the markings checklist pass. Repeatability is incomplete because only three of ten runs are recorded. Use this report to try recording, the live readout, closing and returning.

NAWI/2026/0002, Essae-Teraoka DS-852 (certified)

Class III platform scale, Max 15 kg, e = d = 5 g. All 17 tests recorded and passing, closed, approved and signed, certificate NAWI-C/2026/0001 issued. Use it to see a complete report, the Word and PDF exports, the certificate with its QR code and the verify page. As an approver you can withdraw and reopen it, then close and approve it again to watch a certificate being issued.

NAWI/2026/0003, Essae-Teraoka DS-560 MI (multi-interval, in progress)

Class III retail scale with two partial ranges: $e = 2\text{ g}$ up to 6 kg and $e = 5\text{ g}$ up to 15 kg , opened for verification. Its weighing table shows an e column, because the permissible error, the changeover step and the discrimination weight follow the range each load falls in. Use it to try the partial-range editor on the instrument form and the CSV import (Import CSV button; a template is at `/api/observations-template.csv`).

20 Troubleshooting

Symptom	Cause and fix
localhost:4178 does not open	The server is not running. Run <code>npm start</code> in <code>~/dev/nawi</code> and keep that window open.
“listen EADDRINUSE” on start	Another copy is running. Use it, or stop it, or start with <code>PORT=5000 npm start</code> .
Sent back to the login page	The 12-hour session expired. Sign in again; you return to the same report.
PDF button opens the HTML report instead	No Chrome, Chromium or Edge found on the server. Install one or set <code>CHROME_PATH</code> . Word export always works.
“this action needs the engineer role”	You are signed in as a viewer. Sign in with an account that has the right role.
Record is refused	Read the message under the form; it names the field. Errors are metrological (load above Max, tare above T, missing time) and must be corrected.
A test shows N/A	It does not apply to this instrument's class or type; nothing to record.
Certificate button missing on a passing report	It appears only after approval. Check the outcome panel: every required test must be decided and the instrument must have no classification findings.
Want a fresh database	Stop the server, delete <code>nawi.db</code> , <code>nawi.db-wal</code> and <code>nawi.db-shm</code> , run <code>npm run seed</code> , start again.

21 Glossary

NAWI	Non-automatic weighing instrument: one that needs an operator during weighing (shop scales, platform scales, weighbridges).
Max, Min	Maximum and minimum capacity. Results below Min are not subject to the MPE.
e	Verification scale interval: the interval used for classification and for expressing the MPE.
d	Actual scale interval: what the display steps in. $d \leq e$.
n	Max / e , the number of verification scale intervals. Table 3 limits it per class.

MPE	Maximum permissible error: $\pm 0.5 e$, $\pm 1 e$ or $\pm 1.5 e$ depending on the load band; doubled in service.
Changeover point	The load at which a digital display steps to the next value; used to recover the indication before rounding ($P = I + \frac{1}{2}e - \Delta L$).
E, E₀, E_c	Error of indication; zero error determined before the measurement; corrected error $E_c = E - E_0$, the value compared with the MPE.
Type approval / Verification	Type approval evaluates a design with the full battery; verification checks an individual instrument with the metrological tests.
Initial verification / In service	The two MPE contexts of 3.5.1 and 3.5.2.
Rule set	The JSON file holding every constant of an R 76 edition. Reports remember which one they were judged under.

NAWI TestBench, prototype for Smart India Hackathon 2026, problem statement SIH26035 (Department of Consumer Affairs).
Every constant traces to OIML R 76-1:2006 (E); report layout after R 76-2:2007 (E).